

The Milk Powder Project

- Names (upi) : Bea Cooke (acoo902), Wenting Gu (wgu934), Holly Sutcliffe (hsut050), Christopher Won (cwon491)

Statement/ Specification:

Our project is inspired by a question proposed at MINZ paper (Austral.Mathematical soc., 2018), namely: ‘Can we predict how long we can store milk powders, especially in elevated temperatures and humidities?’

Given the time constraints, we have decided to focus primarily on modelling how the milk powder changed during various temperature settings over a given time. Specifically, we aim to model how heat transfers through a single bag of milk powder. From these models we aim to predict which temperature would be ‘optimal’ for the milk powder to be stored in to prevent as much degradation as possible. This would be useful as temperatures outside the optimal would have more risk for complete degradation, and so may indicate if the bags need to be examined before being sold to the consumer.

Introduction:

The MINZ paper (Austral.Mathematical soc., 2018) outlines a predicted shelf life of milk powder. When milk powder with water activity, $a_w=0.225$ is stored at 25 degrees Celsius (and with relative humidity of 60%), they predict milk powder to have a shelf life of 770 days. We took this information as the minimal optimal conditions for the storage of milk powder, and so with a maximum time to reach the consumer of one year (to account for potential disruption in the supply chain) the consumer should have around 405 days, or just over a year, to consume the product.

Note that there are obviously multiple factors that affect the quality of milk powder, such as moisture content, humidity, and oxygen concentration, as it ships worldwide. However, in this project, we are focusing on how different temperatures influence the quality of milk powder. We want to distinguish the temperature influence from the difference between the powder close to the surface of the bag and deeper in the bag. From this, we aim to infer how the milk powder may have changed over time and identify when milk powder is at risk for degradation (i.e. outside the optimum).

From the MINZ paper, it is known that after a certain level an increase in temperature of the milk powder will lead to lactose crystallization and browning. With our modeling we are trying to avoid the combination of increasing temperature and the glass transition temperature as this leads to all complex inter-particle bridging (and a faster change of rate of the reaction).

To find such results, we use the homogeneous heat diffusion equation for one dimension in space, from Peter J.Olver (2014), that is,

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

where γ refers to thermal diffusivity property, and $u(x,t)$ is the temperature at time t and position x . Note thermal diffusivity, γ , is calculated as

$$\gamma = \frac{\delta}{\rho C_p}$$

where δ is thermal conductivity, ρ is bulk density, and C_p is specific heat capacity. Its unit of measurement is m^2s^{-1} . With regards to milk powder, $\gamma \approx 10^{-7}m^2s^{-1}$ as a result of carrying out numerous calculations using values δ, ρ and C_p of milk powder. For our project, we will set $\gamma = 0.0000007 m^2s^{-1}$ as precise numerical figure, which is also equal to $0.7 mm^2s^{-1}$. Furthermore, a standard 25kg milk powder bag has three dimensions of 830mm by 430mm by 130mm. We have chosen the width (130mm) as the dimension to evaluate this difference. We are assuming that the other properties are constant throughout the milk powder bag for the purpose of our modeling. Through the project, we will combine Olver's heat equation with different boundary conditions and initial conditions to estimate how the milk powder is influenced by the temperature over time, or as a steady-state system.

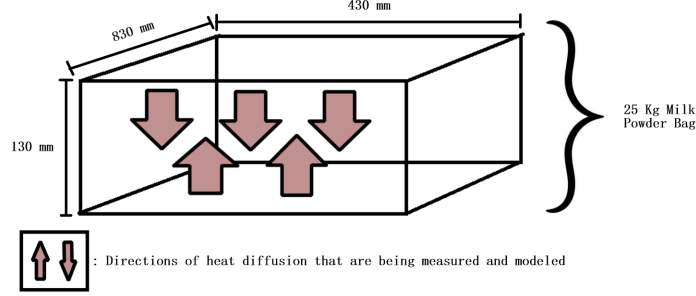


Figure 1: Visual Interpretation

Method

To start, let's model the heat diffusion within milk powder given two types of boundary conditions. This will provide us an idea of how heat diffuses inside and through the milk powder from a general standpoint. Afterwards, we will aim to model the heat diffusion in a pragmatic situation.

Neumann Boundary conditions

We will assume that packaging surrounding the milk powder has no thermal energy flowing in or out, that is, the heat flux is 0. We can derive a solution from the homogeneous heat equation (2014):

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

with boundary conditions and initial condition respectively:

$$\begin{aligned} u_x(t, 0) &= u_x(t, 130) = 0 \\ u(0, x) &= f(x) = 23 + \cos(x) \end{aligned}$$

which gives us the general solution $u(x, t)$,

$$u(t, x) = a_0 + \sum_{n=1}^{\infty} a_n e^{\frac{-(0.7n^2\pi^2t)}{130^2}} \cos\left(\frac{n\pi x}{130}\right)$$

where

$$a_0 = \frac{1}{130} \int_0^{130} 23 + \cos(x) dx$$

and

$$a_n = \frac{2}{130} \int_0^{130} (23 + \cos(x)) \cos\left(\frac{n\pi x}{130}\right) dx$$

for $n=1,2,3,\dots$

When $t=0$,

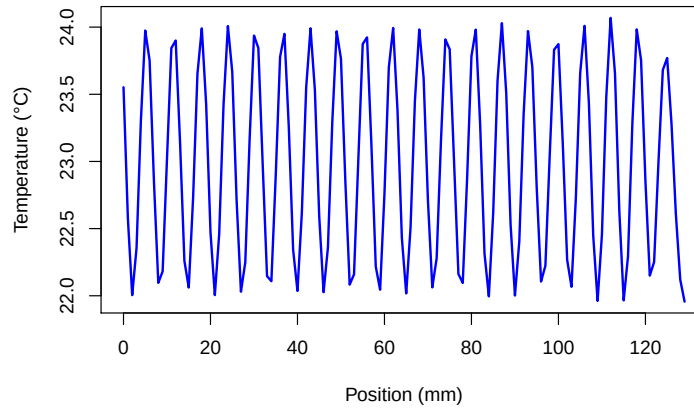


Figure 2: Heat diffusion

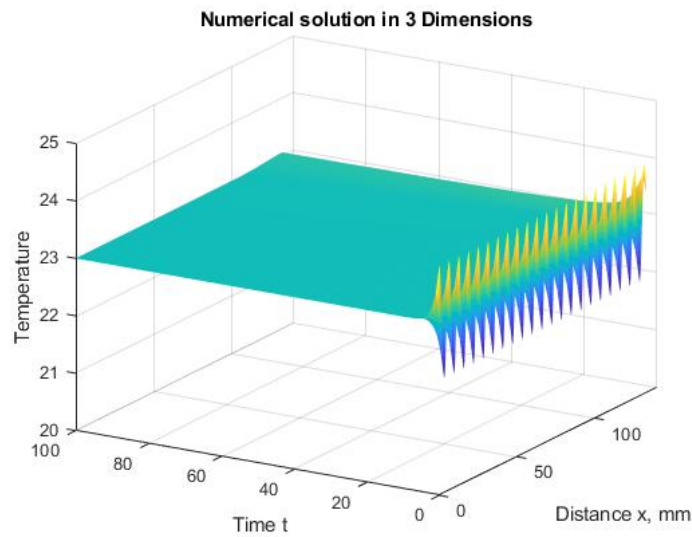


Figure 3: 3-D Interpretation

It can be seen as t increases, we see that the temperature hits an equilibrium of 23°C . We expect to see this since both ends are insulated where the heat flux is 0.

Inhomogeneous Boundary conditions

In this scenario, let's assume the packaged milk powder is face down on one end where there is direct heat flow into one end of the package. Similarly, we will derive a solution from the homogeneous heat equation

(2014):

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

where the boundary conditions and initial condition are respectively:

$$u(t, 0) = 40, u(t, 130) = 35$$

$$u(0, x) = f(x) = 45 + \cos(x) - \frac{10}{130}x$$

and so it follows that the general solution,

$$u(t, x) = 40 + \frac{35 - 40}{130}x + \sum_{n=1}^{\infty} a_n e^{\frac{-(0.7n^2\pi^2t)}{130^2}} \sin\left(\frac{n\pi x}{130}\right)$$

where

$$a_n = \frac{2}{130} \int_0^{130} \left(45 + \cos(x) - \frac{10}{130}x - 40 - \frac{35 - 40}{130}x\right) \sin\left(\frac{n\pi x}{130}\right) dx$$

for $n=1,2,3,\dots$

When $t=0$,

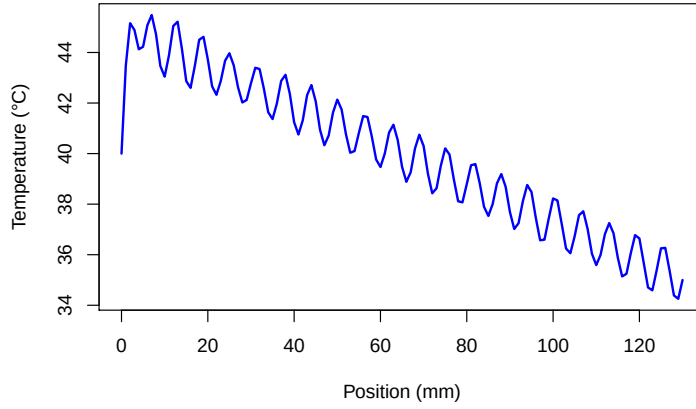


Figure 4: Heat diffusion

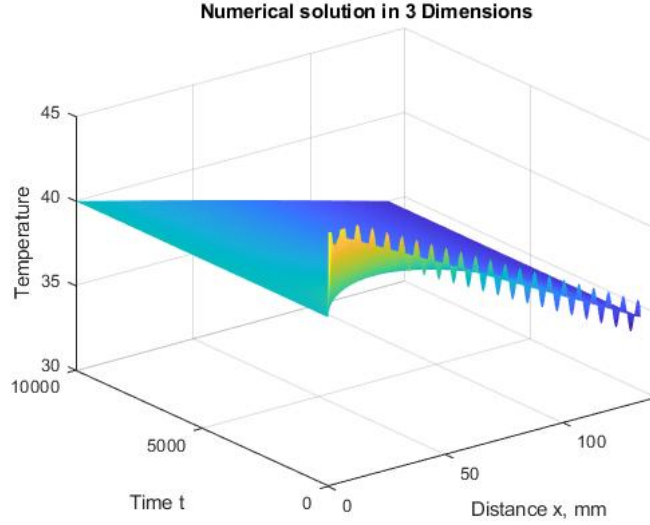


Figure 5: 3-D Interpretation

It appears that there is a sudden spike in temperature increase at the near surface of milk powder where it is bounded by 40°C . We see this happening as the initial temperature that is approximately 5°C , is added. As t increases, the curvature starts to smooth out and the initial temperature distribution converges to an equilibrium distribution of $40 + \frac{35 - 40}{130}x$.

Modeling heat diffusion given a pragmatic event

Typically, 25 kg milk powder is packaged by two types of methods: an inner layer of polyethylene (PE) is glued onto the outer layer of paper bag, or the inner layer of PE is separate from the outer layer (2010). For our project, we will only consider the construction where the inner layer of PE is glued onto the paper bag. It is stated that the thickness of the inner LDPE (low density PE) ranges from 0.076 mm to 0.304 mm in which we will take its average of 0.19 mm as the thickness (2010). We will set the thickness of the outer layer to be 4.81 mm. Hence the thickness of the packaging as a whole is 5 mm.

In order to veers towards correctly modeling the heat diffusion in a realistic situation, let's consider the event where we randomly select a 25 kg packaged milk powder sample in Auckland and send it off to an African country. We will utilize the temperature data on pages M401 and M402 from the MINZ presentation called "Predicting the shelf life of milk powder", as the temperature changes that our sample experiences across approximately 90 days (2018). Note that we are making the assumption that the sample is lying face down on one end throughout the whole period.

The packaged milk powder is initially in Auckland. Thus we define the initial temperature distribution $u(x,0) = f(x)$ under storage conditions in Auckland. We will initially store this sample in an controlled area where the outside temperature is 10.5°C and the inside temperature is 5°C .

It is important to note that the packaging surrounding the milk powder acts as a thermal resistance implying that the packaging causes a certain amount of heat loss before directly heating the milk powder inside. For our sample, lets say that there is a temperature difference of 0.5°C through the 5 mm packaging (this is not accurate as we don't have heat flux sensors to correctly capture the outside and inside temperatures of the packaging). Nevertheless, the outside temperature of the milk powder is 10°C whereas the outside temperature of the packaging (i.e. completely exposed to the environment) is 10.5°C .

Rather than setting inhomogeneous boundary conditions, we will assume a prescribed heat flux, ω (W/mm K), on the top end of the milk powder below the packaging layer (i.e. $u(0) = \omega$), and a constant temperature

on the bottom end, $u(0.130) = 5$, that is, the inside temperature of the packaged milk powder.

Firstly, the heat flux equation is:

$$\omega = -\delta \frac{\partial u}{\partial x} = -\frac{\delta}{\tau}(T_2 - T_1)$$

where δ which is the thermal conductivity, (W/m K), τ which is the thickness in m and T_2, T_1 are inside and outside temperatures ($^{\circ}\text{C}$) respectively. With respect to the milk powder inside; $\delta = 0.61$ (at approximately 10°C), $\tau = 0.130$, and $T_2 = 5$, $T_1 = 10$ (note that $T_1 \neq 10.5$). So we have that,

$$\begin{aligned}\omega &= -\frac{0.61}{0.13}(5 - 10) \\ &= 23.26\end{aligned}$$

[For future reference, ω and δ w.r.t units of millimeters (mm) are $\omega=0.00002326$, and $\delta = 0.00061$ which gives $\frac{\omega}{\delta} = \frac{0.00002326}{0.00061} = 0.038$]

The initial temperature distribution is a steady-state system i.e. independent of time t . Therefore, we derive a simpler equation from the homogeneous heat equation (2014):

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

to this simplified heat equation:

$$\begin{aligned}\gamma \frac{\partial^2 u}{\partial x^2} &= 0 \\ \implies \frac{\partial^2 u}{\partial x^2} &= 0\end{aligned}$$

since γ is non-zero. Then by integration w.r.t x , we have that,

$$\frac{\partial u}{\partial x} = a$$

where a is constant. Also, we know from the heat flux equation that

$$\frac{\partial u}{\partial x} = -\frac{\omega}{\delta}$$

at $x=0$, and so it follows that

$$\begin{aligned}-\frac{\omega_1}{\delta} &= a \\ \implies u(x) &= -\frac{\omega}{\delta}x + c\end{aligned}$$

where c is a constant. We can substitute $u(0.130) = 5$ and have that

$$\begin{aligned}c &= 5 + 0.130\frac{\omega}{\delta} \\ \implies u(x) &= \frac{\omega}{\delta}(0.130 - x) + 5,\end{aligned}$$

that is, $u(x) = u(x,0) = f(x)$ is the initial temperature distribution in Auckland. In other words, We have modeled the heat diffusion into the packaged milk powder in specific temperature conditions as a steady-state system.

Here is a visual representation:

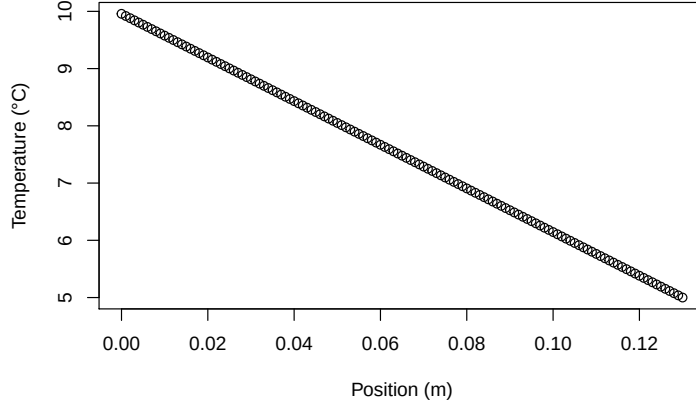


Figure 6: Linear regression of Heat Diffusion

It's quite unusual for heat to travel in a linear manner hence we will model the Figure 6 using a sinusoidal function which may be a more sensible interpretation (Note the units of measurement are in millimeters (mm) for this modeling). We will do this by deriving a general solution from the homogeneous heat equation (2014):

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

where the boundary conditions and initial condition respectively are:

$$\begin{aligned} u_x(t, 0) &= u_x(t, 130) = 0 \\ u(0, x) &= f(x) = 8 + \cos(x) - 0.038x \end{aligned}$$

[Note: We have set Neumann boundary conditions where there is no heat flow in or out of the sample. These boundary conditions were chosen in order to correctly model $u(x)$ in a sinusoidal manner. In the 'Results' section we will discuss having Neumann boundary conditions as being optimal for storing milk powder]

which gives the general solution, $u(t, x)$:

$$u(t, x) = a_0 + \sum_{n=1}^{\infty} a_n e^{\frac{-(0.7n^2\pi^2t)}{130^2}} \cos\left(\frac{n\pi x}{130}\right)$$

where

$$a_0 = \frac{1}{130} \int_0^{130} (8 + \cos(x) - 0.038x) dx$$

$$a_n = \frac{2}{130} \int_0^{130} (8 + \cos(x) - 0.038x) \cos\left(\frac{n\pi x}{130}\right) dx$$

for $n = 1, 2, 3, \dots$

When $t=0$, we have that

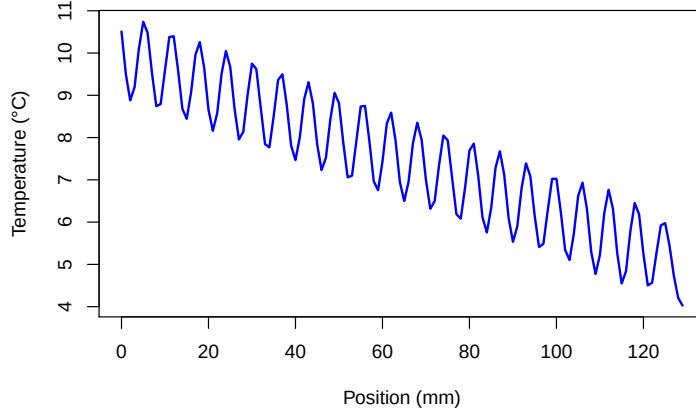


Figure 7: Heat diffusion

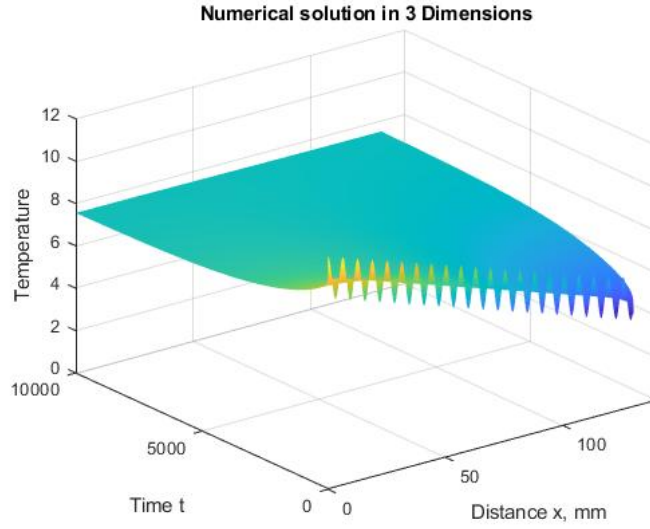


Figure 8: 3-D Interpration

This is an illustration of $u(x)$ when it is $u(x,t)$, that is, not a steady-state system.

Furthermore, we will generalize $u(x)$ by setting the intercept as a variable. In doing so, we can adjust $u(x)$ according to the temperature data of the sample in the African country to model the heat diffusion (2014). That is,

$$u(x) = \frac{\omega}{\delta}(0.130 - x) + p,$$

where p is some temperature.

However, the measured temperature data provided by the MINZ team does not specify how the data was collected; whether it was the surface temperature or the inside temperature of the powder sample. As a consequence, we have no information about the temperature difference and so we are unable to provide true values for T_2 and T_1 . Nevertheless, we will provide the modeling using estimated values to provide an understanding what we may expect to see.

Let's assume the temperature data from the MINZ team is taken as the inside temperature of the powder sample. We can see that the glass transitioning occurs at 50°C with $a_w \approx 0.22$ from Figure 5 of the MINZ paper. We will model the heat diffusion when the sample is at 50°C given that the initial temperature profile is at 26°C (i.e. the starting temperature of the increasing temperature that intersects the glass transition temperature for the first time at day 36.5), using the structure of $u(x)$ as part of the initial condition.

- The following values have been converted with regards to millimeters (mm) for a clearer interpretation, rather than meters (m).

We will carry out the modeling using the homogeneous heat equation:

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

with boundary conditions and initial condition respectively:

$$\begin{aligned} u(t, 0) &= u(t, 130) = 26 \\ u(0, x) &= f(x) = 0.038(130 - x) + 20 \end{aligned}$$

[Note: $0.038 = \frac{\omega}{\delta} = \frac{0.00002326}{0.00061}$ as we converted w.r.t to mm now and similarly, $130 = 1000 * 0.130$].

The general solution $u(t, x)$ follows as,

$$u(t, x) = 26 + \frac{26 - 26}{130}x + \sum_{n=1}^{\infty} b_n e^{\frac{-(0.7n^2\pi^2t)}{130^2}} \sin\left(\frac{n\pi x}{130}\right)$$

where

$$b_n = \frac{2}{130} \int_0^{130} (0.038(130 - x) + 46 - 26) \sin\left(\frac{n\pi x}{130}\right) dx$$

for $n = 1, 2, 3, \dots$

When $t=0$, we have that

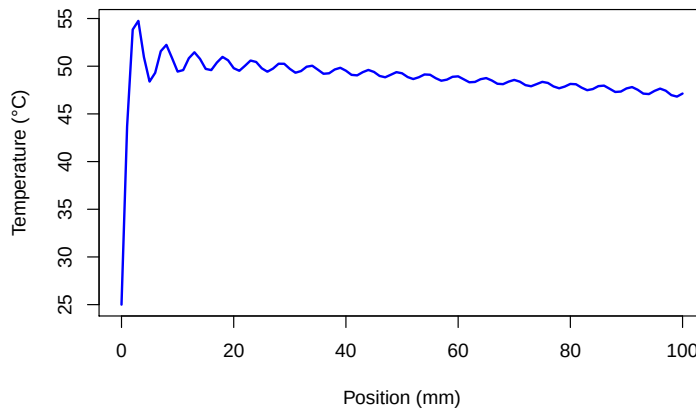


Figure 9: Heat diffusion when sample is at 50°C

- Explanation of Figure 3: The sudden peak at 0 mm until 50°C represents the sudden peak we also see in the temperature data graph. We see a detailed model of the heat diffusion at approximately 50°C

through the sample. In other words, we can visualize the sudden peak as a panned out visual of the heat diffusion, and the decaying sinusoidal function at 50°C as a zoomed-in visual where we can see the extreme details of heat diffusion at a specific temperature. If we zoom into the sudden peak until 50°C that starts at 0 mm, we will see the same decaying sinusoidal function at each unit temperature.

Let's consider another situation where the sample has an initial temperature profile fluctuating around 20°C, and then suddenly the outside temperature rapidly increases to 50°C. We can see this in Figure 4 of the MINZ paper at around day 77 (2018). Similarly, we derive a general solution from the homogeneous heat equation (2014):

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}$$

with boundary conditions and initial condition respectively:

$$\begin{aligned} u(t, 0) &= 20 = u(t, 130) = 19 \\ u(0, x) &= f(x) = 50 + \cos(x) - 0.038x \end{aligned}$$

We have the general solution $u(t, x)$,

$$u(t, x) = 20 + \frac{19 - 20}{130}x + \sum_{n=1}^{\infty} b_n e^{-\frac{(0.7n^2\pi^2t)}{130^2}} \sin\left(\frac{n\pi x}{130}\right)$$

where

$$b_n = \frac{2}{130} \int_0^{130} (50 + \cos(x) - 0.038x - 20 + \frac{1}{130}x) \sin\left(\frac{n\pi x}{130}\right) dx$$

for $n = 1, 2, 3, \dots$

When $t=0$,

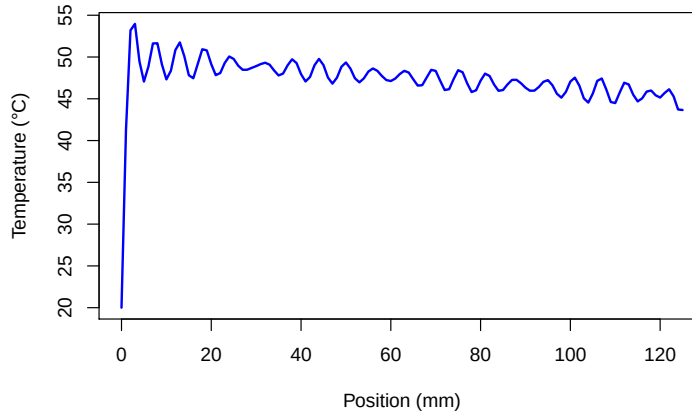


Figure 10: Heat Diffusion of sudden temperature increase

We see an instant jump of temperature to 50°C which is the result of an additional 30°C added onto the initial temperature profile fluctuating at around 20°C as previously mentioned. It is important to note that we are assuming the temperature transitions from 20°C to 50°C in an instant. It would be more reasonable to state that it takes up to a few hours for the temperature to rise 30°C. We however gain an understanding of how the inside temperature could fluctuate immensely in an environment like that of an African country. In the following 'Result' section, we will discuss how we could manage these extreme temperature fluctuations so that we avoid intersecting the glass transition temperature, or unwanted chemical reactions that could negatively affect the quality of milk powder.

Result:

From the modeling that has been carried out in various environmental conditions; we can determine two optimal conditions with respect to temperature in storing milk powder.

It has been found that if the packaging is insulated in a manner where no thermal energy flows in or out, we are able to sustain the initial temperature profile. We saw in both Figure 3 and Figure 8 that t tends to ∞ , the temperature distribution both hit an equilibrium temperature distribution of 23°C and 7.52°C respectively. Given that we maintain $a_w = 0.225$, 60% humidity and an initial temperature profile that ranges to a maximum of 25°C with Neumann boundary condition: $u_x(t, 0) = u_x(t, 130) = 0$, it is possible to maintain its shelf life to at least 770 days. We assume that if there is no heat flux then there is no significant flow of moisture in or out of the packaging that could significantly increase the rate of lactose crystallization as well. Unfortunately such packaging material has not been created for storing milk powder as of date. It would be unrealistic to create such insulation from a multi-wall paper sack with LDPE inner layering (2010). A different type of material would need to be used to accommodate the Neumann boundary conditions in which we leave as a question of interest. Moreover, it is ideal as the location of milk powder becomes less of a concern, knowing that the milk powder will maintain its satisfactory quality despite the temperature fluctuations such as those seen in Figure 4 from the MINZ presentation (2018), over long periods of time.

As it has been previously mentioned, we did not have access to heat flux sensors to correctly measure the heat flux. Given the assumption we do have such access, we could correctly measure the temperature difference from the surface of the milk powder and the bottom end. Furthermore, we can calculate the temperature difference of the packaging which leads us to our second optimal condition. If we are able to calculate the amount of heat loss through the multi-wall paper with LDPE layering (2010), we can make adjustments to the physical make-up of the packaging to increase the heat loss. In other words, the thickness of the paper and LDPE could be corrected to a realistic standard where the outside temperature decreases to a certain extent before reaching the milk powder inside. Undoubtedly, there is a limit to how thick we can extend the materials yet this is much more practical method than having complete insulation all-round. For example, let's consider the glass transitioning at 50°C with $a_w = 0.22$. Thickening the packaging could result in delaying the glass transition temperature as there would be certain amount of heat loss. Also, rather than surrounding the entire milk powder with thick packaging, we could manufacture the packaging to be thicker on the sides that are most likely to be exposed directly to the sun, or any other source of heat. In short, the second optimal condition is to thicken the typical packaging material (2010). This approach however could become costly to carry out as there is a need of additional material. Additionally, it could be less time efficient due to a possible longer manufacturing process. It would be interesting to see the difference in glass transitioning occurrence between two samples where one has 'typical' (2010) packaging and the other has a thicker packaging with the same materials used.

References

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Appendix: Code

- Figure 3,5 and 8 are Matlab code, or otherwise Rmarkdown.

Figure 2:

```
n.sims = 50
integral = numeric(n.sims)
for (i in 1:n.sims) {
  integrand = function(x) {
    (2/130) * (23 + cos(x)) * cos((i * x * pi)/130)
  }
  integral[i] = integrate(integrand, 0, 130)
}
b = c(integral[[1]], integral[[2]], integral[[3]], integral[[4]], integral[[5]],
      integral[[6]], integral[[7]], integral[[8]], integral[[9]], integral[[10]],
      integral[[11]], integral[[12]], integral[[13]], integral[[14]], integral[[15]],
      integral[[16]], integral[[17]], integral[[18]], integral[[19]], integral[[20]],
      integral[[21]], integral[[22]], integral[[23]], integral[[24]], integral[[25]],
      integral[[26]], integral[[27]], integral[[28]], integral[[29]], integral[[30]],
      integral[[31]], integral[[32]], integral[[33]], integral[[34]], integral[[35]],
      integral[[36]], integral[[37]], integral[[38]], integral[[39]], integral[[40]],
      integral[[41]], integral[[42]], integral[[43]], integral[[44]], integral[[45]],
      integral[[46]], integral[[47]], integral[[48]], integral[[49]], integral[[50]])

c = 0.7
t = 0
b = round(b, 4)
n.sims = 130
j = seq(1, 50, by = 1)
Temp = numeric(n.sims)
for (i in 1:n.sims) {
  temp = 23 + sum(((b[j]) * exp(-(c * j^2 * pi^2 * t)/130^2) * cos((j *
    pi * i)/130))))
  Temp[i] = temp
}
x = seq(1, 130, by = 1)
Temps.df = data.frame(x, Temp)
temp.ts = ts(Temps.df$Temp, frequency = 1, start = 0)
plot(temp.ts, ylab = "Temperature (°C)", xlab = "Position (mm)", col = "blue",
      lwd = 2)
```

Figure 3:

```
function [c,f,s] = pdex1pde(x,t,u,dudx) %de
c = 1/0.7;
f = dudx;
s = 0;
end

function u0 = pdex1ic(x) %ic
u0 = cos(x) + 23;
end

function [pl,ql,pr,qr] = pdex1bc(xl,ul,xr,ur,t) %neumann bc
pl = 0;
ql = 1;
pr = 0;
qr = 1;
end
```

Figure 4:

```
n.sims = 50
integral = numeric(n.sims)
for (i in 1:n.sims) {
  integrand = function(x) {
    (2/130) * (cos(x) - (10/130) * x + 45 - 40 + (5/130) * x) * sin((i *
      x * pi)/130)
  }
  integral[i] = integrate(integrand, 0, 130)
}
b = c(integral[[1]], integral[[2]], integral[[3]], integral[[4]], integral[[5]],
  integral[[6]], integral[[7]], integral[[8]], integral[[9]], integral[[10]],
  integral[[11]], integral[[12]], integral[[13]], integral[[14]], integral[[15]],
  integral[[16]], integral[[17]], integral[[18]], integral[[19]], integral[[20]],
  integral[[21]], integral[[22]], integral[[23]], integral[[24]], integral[[25]],
  integral[[26]], integral[[27]], integral[[28]], integral[[29]], integral[[30]],
  integral[[31]], integral[[32]], integral[[33]], integral[[34]], integral[[35]],
  integral[[36]], integral[[37]], integral[[38]], integral[[39]], integral[[40]],
  integral[[41]], integral[[42]], integral[[43]], integral[[44]], integral[[45]],
  integral[[46]], integral[[47]], integral[[48]], integral[[49]], integral[[50]])

c = 0.7
t = 0
b = round(b, 4)
n.sims = 130
j = seq(1, 50, by = 1)
Temp = numeric(n.sims)
for (i in 1:n.sims) {
  temp = 40 - (5/130) * i + sum(((b[j]) * exp(-(c * j^2 * pi^2 * t)/130^2) *
    sin((j * pi * i)/130)))
  Temp[i] = temp
}
x = seq(1, 130, by = 1)
Temps.df = data.frame(x, Temp)
Temps.df = dplyr::add_row(Temps.df, x = 0, Temp = 40, .before = 1)
temp.ts = ts(Temps.df$Temp, frequency = 1, start = 0)
plot(temp.ts, ylab = "Temperature (°C)", xlab = "Position (mm)", col = "blue",
  lwd = "2")
```

Figure 5:

```

function [c,f,s] = pdex1pde(x,t,u,dudx) %de
c = 1/0.7;
f = dudx;
s = 0;
end

function u0 = pdex1ic(x) %ic
u0 = 10+cos(x)-0.038*x;
end

function [pl,ql,pr,qr] = pdex1bc(xl,ul,xr,ur,t) %neumann bc
pl = 0;
ql = 1;
pr = 0;
qr = 1;
end

```

Figure 6:

```

omega = 23.26
delta = 0.61
i = seq(0, 0.13, by = 0.001)
u = ((omega/delta) * (0.13 - i)) + 5
plot(u ~ i, xlab = "Position (m)", ylab = "Temperature (°C)")

```

Figure 7:

```

n.sims = 50
integral = numeric(n.sims)
for (i in 1:n.sims) {
  integrand = function(x) {
    (2/130) * (10 + cos(x) - (0.038 * x)) * cos((i * x * pi)/130)
  }
  integral[i] = integrate(integrand, 0, 130)
}
b = c(integral[[1]], integral[[2]], integral[[3]], integral[[4]], integral[[5]],
integral[[6]], integral[[7]], integral[[8]], integral[[9]], integral[[10]],
integral[[11]], integral[[12]], integral[[13]], integral[[14]], integral[[15]],
integral[[16]], integral[[17]], integral[[18]], integral[[19]], integral[[20]],
integral[[21]], integral[[22]], integral[[23]], integral[[24]], integral[[25]],
integral[[26]], integral[[27]], integral[[28]], integral[[29]], integral[[30]],
integral[[31]], integral[[32]], integral[[33]], integral[[34]], integral[[35]],
integral[[36]], integral[[37]], integral[[38]], integral[[39]], integral[[40]],
integral[[41]], integral[[42]], integral[[43]], integral[[44]], integral[[45]],
integral[[46]], integral[[47]], integral[[48]], integral[[49]], integral[[50]])

c = 0.7
t = 0
b = round(b, 4)
n.sims = 130
j = seq(1, 50, by = 1)
Temp = numeric(n.sims)
for (i in 1:n.sims) {
  temp = 7.52 + sum(((b[j]) * exp(-(c * j^2 * pi^2 * t)/130^2) * cos((j *
pi * i)/130)))
}

```

```

    Temp[i] = temp
}
x = seq(1, 130, by = 1)
Temps.df = data.frame(x, Temp)
temp.ts = ts(Temps.df$Temp, frequency = 1, start = 0)
plot(temp.ts, ylab = "Temperature (°C)", xlab = "Position (mm)", col = "blue",
      lwd = 2)

```

Figure 8:

```

x = linspace(0,130,1000);
t = linspace(0,10000,10000);

m = 0;
sol = pdepe(m,@pdex1pde,@pdex1ic,@pdex1bc,x,t);
u = sol(:,:,1);

h = surf(x,t,u);
title('Numerical solution in 3 Dimensions')
xlabel('Distance x, mm')
ylabel('Time t')
zlabel('Temperature')
zlim([0 12])
xlim([0 140])
set(h,'LineStyle','none')

```

Figure 9:

```

n.sims = 50
integral = numeric(n.sims)
for (i in 1:n.sims) {
    integrand = function(x) {
        (2/130) * (0.038 * (130 - x) + 46 - 26) * sin((i * x * pi)/130)
    }
    integral[i] = integrate(integrand, 0, 130)
}

b = c(integral[[1]], integral[[2]], integral[[3]], integral[[4]], integral[[5]],
      integral[[6]], integral[[7]], integral[[8]], integral[[9]], integral[[10]],
      integral[[11]], integral[[12]], integral[[13]], integral[[14]], integral[[15]],
      integral[[16]], integral[[17]], integral[[18]], integral[[19]], integral[[20]],
      integral[[21]], integral[[22]], integral[[23]], integral[[24]], integral[[25]],
      integral[[26]], integral[[27]], integral[[28]], integral[[29]], integral[[30]],
      integral[[31]], integral[[32]], integral[[33]], integral[[34]], integral[[35]],
      integral[[36]], integral[[37]], integral[[38]], integral[[39]], integral[[40]],
      integral[[41]], integral[[42]], integral[[43]], integral[[44]], integral[[45]],
      integral[[46]], integral[[47]], integral[[48]], integral[[49]], integral[[50]])

c = 0.7
t = 0
b = round(b, 4)
n.sims = 100
j = seq(1, 50, by = 1)
Temp = numeric(n.sims)

```

```

for (i in 1:n.sims) {
  temp = 26 + sum(((b[j]) * exp(-(c * j^2 * pi^2 * t)/130^2) * sin((j *
    pi * i)/130)))
  Temp[i] = temp
}

x = seq(1, 100, by = 1)
Temps.df = data.frame(x, Temp)
Temps.df = dplyr::add_row(Temps.df, x = 0, Temp = 25, .before = 1)
temp.ts = ts(Temps.df$Temp, frequency = 1, start = 0)
plot(temp.ts, ylab = "Temperature (°C)", xlab = "Position (mm)", col = "blue",
  lwd = 2)

```

Figure 10:

```

n.sims = 50
integral = numeric(n.sims)
for (i in 1:n.sims) {
  integrand = function(x) {
    (2/130) * (50 + cos(x) - 0.038 * x - 20 + (1/130) * x) * sin((i *
      x * pi)/130)
  }
  integral[i] = integrate(integrand, 0, 130)
}

b = c(integral[[1]], integral[[2]], integral[[3]], integral[[4]], integral[[5]],
  integral[[6]], integral[[7]], integral[[8]], integral[[9]], integral[[10]],
  integral[[11]], integral[[12]], integral[[13]], integral[[14]], integral[[15]],
  integral[[16]], integral[[17]], integral[[18]], integral[[19]], integral[[20]],
  integral[[21]], integral[[22]], integral[[23]], integral[[24]], integral[[25]],
  integral[[26]], integral[[27]], integral[[28]], integral[[29]], integral[[30]],
  integral[[31]], integral[[32]], integral[[33]], integral[[34]], integral[[35]],
  integral[[36]], integral[[37]], integral[[38]], integral[[39]], integral[[40]],
  integral[[41]], integral[[42]], integral[[43]], integral[[44]], integral[[45]],
  integral[[46]], integral[[47]], integral[[48]], integral[[49]], integral[[50]])

```

```

c = 0.7
t = 0
b = round(b, 4)
n.sims = 125
j = seq(1, 50, by = 1)
Temp = numeric(n.sims)
for (i in 1:n.sims) {
  temp = 20 - (1/130) * i + sum(((b[j]) * exp(-(c * j^2 * pi^2 * t)/130^2) *
    sin((j * pi * i)/130)))
  Temp[i] = temp
}

x = seq(1, 125, by = 1)
Temps.df = data.frame(x, Temp)
Temps.df = dplyr::add_row(Temps.df, x = 0, Temp = 20, .before = 1)
temp.ts = ts(Temps.df$Temp, frequency = 1, start = 0)
plot(temp.ts, ylab = "Temperature (°C)", xlab = "Position (mm)", col = "blue",

```

```
lwd = 2)
```